

DevOps Lab Preparation

Program8: Organize kubernetes Pods using Labels and Namespaces

Project Structure:

Program9/

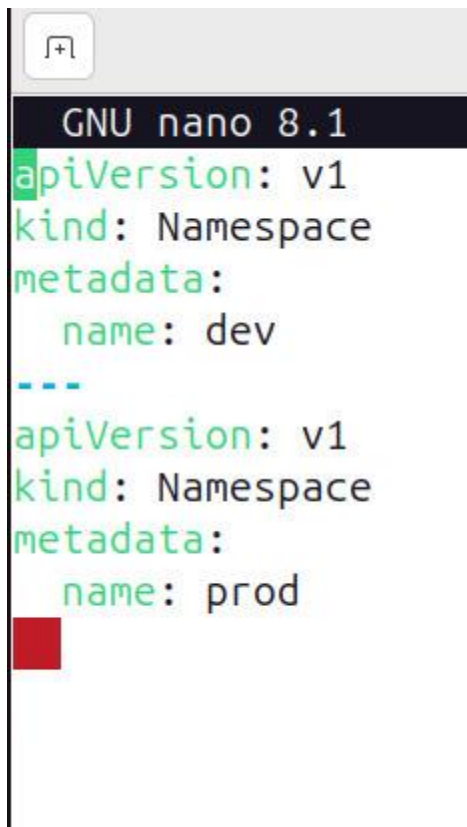
→namespaces.yaml

→pods.yaml

Steps for execution:

Step 1: Start minikube and check the status

Step 2: Create namespaces.yaml file with the following content



```
GNU nano 8.1
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 3: Create pods.yaml file

```

GNU nano 8.1
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
      - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
      - containerPort: 80
---

```

```

GNU nano 8.1
      ports:
      - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
      - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
      - containerPort: 80

```

Step 4: Apply the files

Kubectl apply -f namespaces.yaml

Kubectl apply -f pods.yaml

Step 5: Verify the pods with namespaces

Kubectl get pods -n dev

Kubectl get pods -n prod

Step 6: Get all of the pods

Step 7: Verify pods with labels

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab8
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab9
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          16s
frontend-pod  0/1     ContainerCreating   0          16s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          24s
frontend-pod  1/1     Running   0          24s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          26s
frontend-pod  1/1     Running           0          26s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          31s
frontend-pod  1/1     Running   0          31s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods
No resources found in default namespace.
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n prod -l app:backend
Error from server (BadRequest): Unable to find "/v1, Resource=pods" that match label selector "app:backend", field selector "": unable to parse requirement: <nil>: Invalid value: "app:backend": name part must consist of alphanumeric characters, '-', '_' or '.', and must start and end with an alphanumeric character (e.g. 'MyName', or 'my.name', or '123-abc', regex used for validation is '([A-Za-z0-9](-|_|\.)?[A-Za-z0-9])')
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n dev -l app:backend
Error from server (BadRequest): Unable to find "/v1, Resource=pods" that match label selector "app:backend", field selector "": unable to parse requirement: <nil>: Invalid value: "app:backend": name part must consist of alphanumeric characters, '-', '_' or '.', and must start and end with an alphanumeric character (e.g. 'MyName', or 'my.name', or '123-abc', regex used for validation is '([A-Za-z0-9](-|_|\.)?[A-Za-z0-9])')
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n prod -l app:backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          113s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n dev -l app:frontend
Error from server (BadRequest): Unable to find "/v1, Resource=pods" that match label selector "app:frontend", field selector "": unable to parse requirement: <nil>: Invalid value: "app:frontend": name part must consist of alphanumeric characters, '-', '_' or '.', and must start and end with an alphanumeric character (e.g. 'MyName', or 'my.name', or '123-abc', regex used for validation is '([A-Za-z0-9](-|_|\.)?[A-Za-z0-9])')
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          2m8s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          2m12s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          2m16s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab9$
```